

Math Exam-03 — Attempt Overview

Units 4, 5, 7 · Factorization, Linear Equations & Inequalities, Coordinate Geometry · Seemab · 2026-04-22

SINGLE ATTEMPT · 6TH CONSECUTIVE PERFECT MCQ SCORE

60.5 / 65

93.1% · Grade A+ · Section A MCQs 12/12

12 / 12

Section A — MCQs
100%

29.5 / 33

Section B — Short
89.4%

19 / 20

Section C — Long
95.0%

What Went Well

- **MCQs 12/12 clean** — every factoring pattern ($x^4 + 4y^4$, sum of cubes, HCF identity), every inequality trick (empty solution set for $\sqrt{(5x)} = -10$), every coordinate-geometry shortcut (distance 13, midpoint (4,7)) landed. Six consecutive perfect MCQ scores now.
- **HCF / LCM procedure clean** — Q.2(iii) and Q.2(iv)(or): correct coefficient HCF/LCM + min/max power rule on variables, $6xy^2$ and $24a^2b^3$ both spot on.
- **Linear equation and absolute-value equation** — Q.2(vi)(or) solved $5x-2-x = 4-3x-27 \rightarrow x = -3$ clean, and Q.2(vii) $|2x+5| = 11$ handled both cases cleanly to get $\{3, -8\}$.
- **Inequality solution set in formal notation** — Q.2(viii)(or): $8x+9 < 6x-7 \rightarrow x < -8$, written as $\{x \mid x \in \mathbb{R} \wedge x < -8\}$. Textbook-standard set-builder form.
- **Distance formula** — Q.2(ix): $d = \sqrt{(3^2 + 4^2)} = 5$. Pythagorean triple recognised.
- **Collinearity via $AB + BC = AC$** — Q.2(xi): computed all three distances, showed $\sqrt{5} + \sqrt{5} = 2\sqrt{5}$. Correct strategy, correct arithmetic.
- **Complex factorization by substitution** — Q.3(i): paired $(x+5)(x+3)$ with $(x+2)(x+6)$ (equal sums = 8), let $a = x^2 + 8x$, factored $a^2 + 27a + 92 = (a+4)(a+23)$, substituted back. Full 5/5. This is the hardest question on the paper.
- **Square root by long division** — Q.3(ii): laid out the tableau correctly, built partial roots $x^2 \rightarrow x^2 + 4x \rightarrow x^2 + 4x + 2$, remainder 0. Division method executed cleanly — the textbook algorithm that many students fumble.
- **Square proof from coordinates** — Q.3(iv): computed all four sides $= \sqrt{29}$ and both diagonals $= \sqrt{58}$. All 8 distances correct on the first pass.

Where Marks Were Lost

- **Q.2(i)(or)(a) — sign error in grouping** — 1.0 mark lost. Split was correct ($x^2 - 5x + x - 5$) but the grouping line came out as $x(x-5) - 1(x-5)$ instead of $x(x-5) + 1(x-5)$, producing $(x-5)(x-1)$. Expand-to-check shows $(x-5)(x-1) = x^2 - 6x + 5$, not the starting polynomial. The fix is the sign of the second group: when the last term in the split is positive, the grouping factor must also be positive. **Correct answer: $(x-5)(x+1)$.**
- **Q.2(ii)(or) — difference-of-cubes formula applied loosely** — 2.0 marks lost. Three parallel errors: (i) first factor written as $(2x - 5)$ instead of $(2x - 5y)$ — the y was dropped; (ii) $(2x)^2$ written as $2x^2$ instead of $4x^2$; (iii) $(5y)^2$ written as $5y^2$ instead of $25y^2$. All three are squaring mistakes on grouped symbols — a parentheses discipline issue. **Correct answer: $(2x - 5y)(4x^2 + 10xy + 25y^2)$.**
- **Q.2(v) — missing $\pm$ on square root** — 0.5 mark lost. Perfect-square factorization $(2x+3y)^2$ was correct, but the final square root should be $\pm(2x + 3y)$, not just $(2x + 3y)$. Every square root has two values; the $\pm$ is non-negotiable.
- **Q.3(iii) — no number-line representation** — 1.0 mark lost. Solution set $\{x \mid x \in \mathbb{R} \wedge x > 2 \text{ OR } x < 1\}$ was correct, but the marking scheme awards 1 mark for the number-line sketch with open circles at 1 and 2 and arrows going outward. Worth adding as a habit for every compound-inequality question.

Item-by-Item Marking

Section A — MCQs (12 / 12)

#	Topic	Her Answer	Key	Verdict
1	Factorize $x^4 + 4y^4$	A	A	✓
2	Factors of $y^2 - 7y + 12$	A	A	✓
3	HCF of $18a^3b^4c^5$ and $24a^2b^5c^3$	A	A	✓
4	HCF $\times$ LCM of two polynomials	C	C	✓
5	$\sqrt{(9a^2 + 12ab + 4b^2)}$	A	A	✓
6	Factor $a^3 + 27b^3$	A	A	✓
7	Linear equation in one variable	C	C	✓
8	$ x + 1 = 2$	A	A	✓
9	$\sqrt{(5x)} = -10$	B	B	✓ (empty set)
10	Distance A(-1, 3) to B(4, 15)	B	B	✓
11	Minimum sides of polygon	B	B	✓
12	Midpoint of (2,4) & (6,10)	C	C	✓
Section A Total				12 / 12

Section B — Short Questions (29.5 / 33)

Q	Chose	Topic	Verdict	Score
2(i)	OR	Factorize $x^2 - 4x - 5$; $2x^2 + 9x + 4$	Partial — (a) sign slip; (b) correct	2.0 / 3
2(ii)	OR	Factorize $8x^3 - 125y^3$	Partial — formula ok, squaring botched	1.0 / 3
2(iii)	Main	HCF of $12x^2y^3$ and $18xy^2$	Correct — $6xy^2$	3 / 3
2(iv)	OR	LCM of $6a^2b$ and $8ab^3$	Correct — $24a^2b^3$	3 / 3
2(v)	Main	$\sqrt{(4x^2 + 12xy + 9y^2)}$	Missing $\pm$ sign	2.5 / 3
2(vi)	OR	Solve $5x - 2 - x = 4 - 3x - 27$	Correct — $x = -3$	3 / 3
2(vii)	Main	$ 2x + 5 = 11$	Correct — $\{3, -8\}$	3 / 3
2(viii)	OR	$8x + 9 < 6x - 7$	Correct — $\{x \mid x < -8\}$	3 / 3
2(ix)	Main	Distance (-2,1) to (1,5)	Correct — 5 units	3 / 3
2(x)	Main	Midpoint (2,5) & (6,9)	Correct — (4, 7)	3 / 3
2(xi)	Main	Collinearity of A(3,2), B(4,4), C(5,6)	Correct — $AB + BC = AC = 2\sqrt{5}$	3 / 3
Section B Total				29.5 / 33

Section C — Long Questions (19 / 20)

Q	Chose	Topic	Verdict	Score
3(i)	Main	Factorize $(x+5)(x+3)(x+2)(x+6) - 88$	Full substitution method, $(x^2 + 8x + 4)(x^2 + 8x + 23)$	5 / 5
3(ii)	Main	$\sqrt{(x^4 + 8x^3 + 20x^2 + 16x + 4)}$ by division	Clean long division, $\pm(x^2 + 4x + 2)$	5 / 5
3(iii)	Main	Compound inequality $2x + 3 > 7$ OR $4x - 1 < 3$	Solution set correct; no number line	4 / 5

3(iv)	Main	A(-4,2), B(1,4), C(3,-1), D(-2,-3) is a square	All 4 sides = $\sqrt{29}$, both diagonals = $\sqrt{58}$	5 / 5
Section C Total				19 / 20

Key Techniques Seemab Demonstrated

Technique	Where Shown
Sum/difference of cubes identity recognition	A-Q6, Q.2(ii) OR
HCF / LCM via prime-factor trees	Q.2(iii), Q.2(iv) OR
Perfect-square identity $a^2 + 2ab + b^2 = (a+b)^2$	A-Q5, Q.2(v)
Solving absolute-value equation by case split	A-Q8, Q.2(vii)
Empty solution set when $\sqrt{\cdot}$ = negative	A-Q9
Set-builder notation for inequality solutions	Q.2(viii) OR, Q.3(iii)
Distance formula $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$	A-Q10, Q.2(ix), Q.2(xi), Q.3(iv)
Midpoint formula $((x_1+x_2)/2, (y_1+y_2)/2)$	A-Q12, Q.2(x)
Collinearity via $AB + BC = AC$	Q.2(xi)
Substitution factoring (let u = quadratic expression)	Q.3(i)
Square-root by polynomial long division	Q.3(ii)
Square proof: all sides equal + diagonals equal	Q.3(iv)

Pattern Check Against Previous Attempts

Parentheses discipline is the one recurring theme. The two biggest deductions on this paper — Q.2(i) (a) sign slip and Q.2(ii)(or) squaring errors — both trace to the same root cause: when manipulating a grouped symbol like $(2x)$ or $-1(x-5)$, the grouping wasn't fully respected. On Q.2(ii), $(2x)^2$ got collapsed to $2x^2$ (should be $4x^2$), and $(5y)^2$ to $5y^2$ (should be $25y^2$). Suggested habit: when a coefficient sits inside parentheses with a variable and gets squared, square both. Write it out as $(2x)(2x) = 4x^2$ the first few times until it's automatic.

The rest of the paper is strong. Section C's Q.3(i) and Q.3(ii) are two of the hardest question-types in the whole syllabus (chained substitution factoring, and square-root by polynomial long division). Both were executed cleanly. This is A+ level work — the deductions are isolated and fixable, not systemic.

The MCQ streak extends to 6. Six back-to-back 12/12s across Phys-01, Phys-02, Chem-01, Chem-02, Math-01, Math-03. The only MCQ miss in Seemab's full record remains the single Biology MCQ on the very first exam (2026-04-03).

Position in Track Record

This is Seemab's **7th exam** and her **2nd math exam**.

#	Subject	Chapters	Date	Score	%	MCQ
1	Biology	Ch 1-6	2026-04-03	52 / 65	80.0%	11/12
2	Physics	Ch 1-2	2026-04-07	63.5 / 65	97.7%	12/12
3	Physics	Ch 3-4, 9	2026-04-10	62 / 65	95.4%	12/12
4	Chemistry	Ch 1-2	2026-04-12	64 / 65	98.5%	12/12
5	Chemistry	Ch 3-5	2026-04-17	59.5 / 65	91.5%	12/12
6	Math	Units 1-2	2026-04-18	65 / 65	100%	12/12

7	Math	Units 4, 5, 7	2026-04-22	60.5 / 65	93.1%	12/12
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New running average: 93.7% (426.5 / 455 across 7 exams). MCQ record is now **83 / 84 = 98.8%**. Math two-exam average: **96.5%** — still Seemab's strongest subject.

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